

Regression Analysis (I)

Solutions to Midterm Exam.

April 17, 2017 (Prof. Kim Choongrak)

1.(30) Let $\mathbf{A}$ be $n \times n$ symmetric matrix with eigenvalues $\lambda_1, \dots, \lambda_n$. Using the spectral decomposition theorem, express the determinant of $\mathbf{A}^{-1}$ in terms of the eigenvalues of $\mathbf{A}$. (Ans.) $\mathbf{A} = \mathbf{\Gamma} \mathbf{\Lambda} \mathbf{\Gamma}^T \Rightarrow \mathbf{A}^{-1} = \mathbf{\Gamma} \mathbf{\Lambda}^{-1} \mathbf{\Gamma}^T \Rightarrow |\mathbf{A}^{-1}| = |\mathbf{\Gamma} \mathbf{\Lambda}^{-1} \mathbf{\Gamma}^T| = |\mathbf{\Lambda}^{-1}| = \prod_{i=1}^n \lambda_i^{-1}$.

2.(90) Let $\mathbf{X}$ be $n \times p$ matrix ($p < n$) with $\text{rank}(\mathbf{X}) = p$, and let $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$.

(1)(30) Find $\text{tr}(\mathbf{H})$. (Ans.) $\text{tr}(\mathbf{H}) = \text{tr}(\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T) = \text{tr}((\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X}) = \text{tr}(\mathbf{I}_p) = p$.

(2)(30) Show that $\mathbf{H}$ is idempotent. (Ans.) $\mathbf{H}^2 = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \mathbf{H}$.

(3)(30) Find the rank of $\mathbf{I} - \mathbf{H}$. (Ans.) $r(\mathbf{I} - \mathbf{H}) = n - r(\mathbf{H}) = n - p$ because the rank of idempotent matrix is the same as the trace.

3.(30) Let $\mathbf{x}$ be an n -random vector with p.d. $\text{Cov}(\mathbf{x})$. Show that $\text{Var}(\mathbf{a}^T \mathbf{x}) > 0$ if $\mathbf{a} \neq \mathbf{0}$ is constant vector. (Ans.) $\text{Var}(\mathbf{a}^T \mathbf{x}) = \mathbf{a}^T \mathbf{\Sigma} \mathbf{a} > 0$ because $\text{Cov}(\mathbf{x}) = \mathbf{\Sigma}$ is p.d.

4.(30) Let $\mathbf{A}$ be $n \times n$ matrix, $\mathbf{b}$ be p -vector, and $\mathbf{x}$ be p -vector. Compute $\frac{\partial^2 Q(\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}^T}$, where $Q(\mathbf{x}) = (\mathbf{b} - \mathbf{A}\mathbf{x})^T (\mathbf{b} - \mathbf{A}\mathbf{x})$. (Ans.) $Q(\mathbf{x}) = (\mathbf{b} - \mathbf{A}\mathbf{x})^T (\mathbf{b} - \mathbf{A}\mathbf{x}) = \mathbf{b}^T \mathbf{b} - 2\mathbf{x}^T \mathbf{A}^T \mathbf{b} + \mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x} \Rightarrow \frac{\partial Q(\mathbf{x})}{\partial \mathbf{x}} = 2\mathbf{A}^T \mathbf{b} + 2\mathbf{A}^T \mathbf{A} \mathbf{x} \Rightarrow \frac{\partial^2 Q(\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}^T} = 2\mathbf{A}^T \mathbf{A}$.

5.(100) Let $X_1, \dots, X_n$ be iid $N(\mu, \sigma^2)$, where μ and σ are unknown. Also, let $\bar{X} = \sum X_i/n$, $s^2 = \sum (X_i - \bar{X})^2/(n-1)$.

(1)(30) Find $E((\bar{X}s)^2)$. (Ans.) $E((\bar{X}s)^2) = E[\bar{X}^2]E[s^2]$ because $\bar{X}$ and s^2 are independent. Now, $E[\bar{X}^2] = \text{Var}(\bar{X}) + E^2[\bar{X}] = \frac{\sigma^2}{n} + \mu^2$, and $E(s^2) = \sigma^2$. Hence, $E((\bar{X}s)^2) = \sigma^2(\frac{\sigma^2}{n} + \mu^2)$.

(2)(30) Find $100(1-\alpha)\%$ C.I. for σ . (Ans.) Since $(n-1)s^2/\sigma^2 \sim \chi^2(n-1)$, we have $1-\alpha = P\left[\chi_{1-\alpha/2}^2 < \frac{(n-1)s^2}{\sigma^2} < \chi_{\alpha/2}^2\right]$, the $100(1-\alpha)\%$ C.I. for σ^2 is $\left(\frac{(n-1)s^2}{\chi_{\alpha/2}^2}, \frac{(n-1)s^2}{\chi_{1-\alpha/2}^2}\right)$. Therefore, the $100(1-\alpha)\%$ C.I. for σ is $\left(\sqrt{\frac{(n-1)s^2}{\chi_{\alpha/2}^2}}, \sqrt{\frac{(n-1)s^2}{\chi_{1-\alpha/2}^2}}\right)$.

(3)(40) Find the distribution of $n(\bar{X} - \mu)^2/s^2$. (Ans.) We showed that (done in class) $\sqrt{n}(\bar{X} - \mu)/s \sim t(n-1)$. Hence, $n(\bar{X} - \mu)^2/s^2 \sim F(1, n-1)$.

6.(40) Let $X_1, \dots, X_n$ be iid $N(0, \theta)$. Compute the MLE of θ . (Ans.) $L(\theta) = (2\pi\theta)^{-n/2} \exp(-\frac{\sum X_i^2}{2\theta}) \Rightarrow l(\theta) = -\frac{n}{2} \log((2\pi\theta)) - \frac{\sum X_i^2}{2\theta} \Rightarrow \frac{\partial l(\theta)}{\partial \theta} = -\frac{n}{2} \frac{1}{\theta} + \frac{\sum X_i^2}{2\theta^2} = 0 \Rightarrow \hat{\theta} = \sum X_i^2/n$.

7.(80) Consider a simple linear regression model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$, $i = 1, \dots, n$. Assume that ϵ_i 's are independent and $E(\epsilon_i) = 0$, $E(\epsilon_i^2) = \sigma^2$. Let $\hat{\beta}_0$ and $\hat{\beta}_1$ be LSEs of β_0 and β_1 , respectively. (Ans.) See the text book.

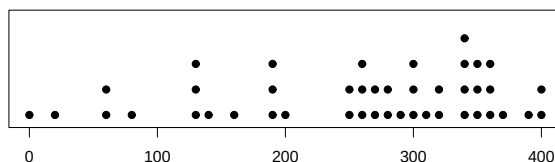
(1)(40) When we write $\hat{\beta}_0 = \sum_{i=1}^n w_i^* Y_i$, find w_i^* .

(2)(40) Using the result of (1), compute $Var(\hat{\beta}_0)$.

Summary Statistics

$n = 44$, sample mean = 255.7

Scores for Midterm Exam. in Regression Analysis (I), 2017



Top Seat : 남승현, 유가영